香港中文大學（深圳）
The Chinese University of Hong Kong, Shenzhen

Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	CSC3050
Course title (English)	Computer Architecture
Course title (Chinese)	计算机体系结构
Units	3
Language of Instruction	English
Description (English)	Introduction of the major components of a computer system, including instruction set architecture (ISA), CPU datapath, control unit design, pipelining, memory hierarchy, peripheral interfaces, and multiprocessor architecture. RISC-V architecture will be used as a reference architecture.
Description (Chinese)	介绍计算机系统的主要组件，包括指令集架构 （ISA）、计算机的整数和浮点运算，CPU 数据路径、控制单元设计、流水线、内存层次结构（缓存）、外设接口和多处理器架构。RISC-V架构将用作参考架构。

2. Prerequisites / Co-requisites

A. Prerequisites

CSC3002 C/C++ Programming

B. Co-requisites

3. Learning Outcomes

- Read, write, and debug RISC-V assembly language code
- Design and implement a pipelined RISC-V RV32I simulator
- Design and implement the Vector Extension (RVV1.0) on top of the RISC-V RV32I simulator

- Design and implement a multi-level cache simulator
- Understand how data (both text and numbers) and instructions are represented in computers
- Understand how integer and floating-point arithmetic work

4. Course syllabus

Goals :
1. Understand the principles of Computer Architecture and their implementation details.
2. Use software to design a simulator of a pipelined RISC-V RV32I processor.
3. Understand cache performance through the implementation and simulation of a multi-level cache simulator.

Textbook :
Computer Organization and Design: The Hardware/Software Interface (RISC-V Edition), 2018

References:
1. Documentation: RISC-V Instruction Set Manual.
2. Computer Systems: A Programmer's Perspective 3e

Contents:
- Introduction
- Digital Logic Review
- Instruction Set Architecture
- Arithmetic and Logic Unit
- Datapath
- Pipelining
- Memory and Cache
- Virtual-Memory
- Virtual Machines
- Instruction Level Parallelism, Thread Level Parallelism, Data Level Parallelism, Memory Level Parallelism

Grading:
Class participation - 10%
Quizzes – 20%
Programs – 45%
Final – 25%

5. Assessment Scheme

Component/ method	% weight
Class Participation	10
Quizzes	20
Programming Assignments	45
Final Examination	25

6. Grade descriptor

General

Grade	Description
A	Demonstrates the ability to design and implement a pipelined computer processor simulator with techniques to reduce performance penalty of hazard avoidance. Demonstrates the ability to design and implement a multi-level cache simulator, including performance enhancement features. Has the ability to efficiently read, write, and debug assembly language programs independently. Understand the processor performance tradeoffs.
A-	Demonstrates the ability to design and implement a pipelined computer processor simulator. Demonstrates the ability to design and implement a multi-level cache simulator. Has the ability to efficiently read, write, and debug assembly language programs independently. Understand the processor performance tradeoffs.
B	Demonstrates the ability to design and implement a computer processor simulator. Demonstrates the ability to design and implement a cache simulator. Has the ability to efficiently read, write, and debug assembly language programs independently. Understand the processor performance tradeoffs.
C	Demonstrates the ability to implement a computer processor simulator in C/C++. Demonstrates the ability to design and implement a cache simulator. Has the ability to efficiently read, write, and debug assembly language programs independently. Understand the processor performance tradeoffs.
D	Demonstrates the ability to implement a computer processor simulator and a cache simulator in C/C++. Has the ability to read, write, and debug assembly language programs.
F	Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.

7. Feedback for evaluation

- Formal course and teaching evaluation
- In-class questions, wechat messages, emails, communication via course website

(blackboard)
- Informal feedback to instructor and TA via office hours
- Departmental retreat and program review

8. Reading

Textbook

1.Computer Organization and Design: The Hardware/Software Interface (RISC-V Edition), 0128122757, Patterson and Hennessy, Morgan Kaufmann, 2ND, 2017, United States

2.Computer Systems: A Programmer's Perspective 3e, 978-1-488-67207-1, Bryant and O'Hallaron, 3RD, 2016, United States

9. Course components

Activity	Hours/week
Lecture	3
Readings & Assignments	8
Tutorial	1

10. Indicative teaching plan

Week	Content/ topic/ activity
001	Introduction
002	Digital Logic Basics
003	Instruction Set Architecture, RISC vs CISC, RISC-V architecture
004	Arithmetic for computers
005	DataPath Design: single cycle and multi-cycle implementation
006	Pipelining
007	Hazards reduction in pipelined processors
008	Memory Organization and Memory Hierarchy
009	Cache
010	High performance cache organizations
011	Virtual Memory and Virtual Machines
012	ILP, TLP, DLP, and MLP
013	Multi-threading and Multi-cores
014	Review

11. Any other information

12. Version date

Version number	5
As of (date)	2024/11/18